Duale Hochschule Baden-Württemberg Stuttgart

Springer
Autonomous Intelligent Systems

22.01.2026

Paper Resubmission

Dear Yunfeng Hu,

We sincerely appreciate the opportunity to revise our manuscript, *Tiny Neural Networks for Multi-Object Tracking in a Modular Kalman Framework*, for Autonomous Intelligent Systems. We are grateful for the constructive feedback provided by the reviewers. Their insightful comments have helped us significantly improve the clarity and depth of our work.

Detailed response to the reviewers:

Reviewer #1: This paper presents a study on Multi-Object Tracking (MOT), proposing a modular and production-oriented method that integrates a compact neural network (NN) into a Kalman filter-based MOT pipeline. The approach includes the SPENT (Single-Prediction Network), which predicts per-track states and replaces heuristic motion models used by the Kalman Filter (KF); the SANT (Single-Association Network), which assigns incoming sensor objects to existing tracks without relying on heuristic distance or association metrics; and the MANTa (Multi-Association Network), which jointly associates multiple sensor objects to multiple tracks in a single step. The method maintains modularity, interpretability, and real-time suitability for embedded automotive driver assistance systems, demonstrating novelty and practical engineering significance. However, several issues require further resolution prior to publication.

1. The state prediction equation is replaced by SPENT. How is the measurement equation in the Kalman Filter handled in this context?

→ For our Kalman set-up, the predicted states from the SPENT network have the same representation as the sensor objects (relative coordinates to ego within a five-dimensional state vector). Thus, here the H Matrix in the measurement equation is a unit matrix and we set the noise vector to zero. We extended the manuscript to make this clear for the reader (see IV, B.).

2. To simulate more realistic sensor data, artificial noise is introduced. Is this artificial noise reasonable? Please analyze whether it is simple Gaussian white noise or a more complex type. This aspect requires more detailed explanation.

→ In the revised manuscript, we explicitly clarify the noise model used for generating synthetic sensor objects. The applied noise is additive, zero-mean, independent Gaussian noise, scaled relative to each state component (maximum 3%). This corresponds to a scaled white Gaussian noise model without temporal or cross-dimensional correlations.

DHBW Stuttgart
Rotebühlstr. 133
70197 Stuttgart
Telefon + 49 . 711 . 18 49 - 632
Telefax + 49 . 711 . 18 49 - 719

DHBW Stuttgart
Campus Horb
Florianstraße 15
72160 Horb
Telefon + 49 . 74 51 . 521 - 0
Telefax + 49 . 74 51 . 521 - 111

www.dhbw-stuttgart.de

We further discuss the rationale and limitations of this simplified noise model, emphasizing that it is intended to introduce controlled measurement uncertainty rather than to fully emulate real sensor error characteristics. Additional explanations were added to Section IV-B.

3. Three networks were trained in the study. Are these networks interconnected, and how is the overall effectiveness of these networks ensured in practical application?

→ The underlying philosophy of our approach is to conserve modularity and interpretability as much as possible. Thus, each of the three networks are trained independently of the others. This also means that their performance can be assessed independently. The prediction network SPENT is evaluated through the Root Mean Square Error of the prediction positions (old Table I, now Fig. 7) and the two association networks (SANT and MANTa) are evaluated through their association accuracy (old Table II, now Fig. 8). This induvial testing ensures effectiveness and the possibility to localize areas of improvement. Since this is a core development guideline for our work, we now emphasize this clearer in our abstract and the conclusion.

4. The validation was conducted only on the KITTI dataset, which is often considered limited. Further validation on other datasets should be performed to demonstrate the method's transferability.

→ We thank the reviewer for this valuable suggestion. We agree that evaluating the proposed approach on additional datasets would further strengthen the assessment of transferability. In the present work, we deliberately focus on the KITTI dataset, as it is a well-established benchmark for automotive multi-object tracking and widely used for evaluating both classical and learning-based MOT approaches. Our primary objective is to demonstrate the feasibility, modularity, and system-level integration of the proposed neural components (SPENT, SANT, and MANTa) within a Kalman filter-based Tracking-by-Detection framework, rather than to maximize benchmark coverage. The modular design of our framework allows straightforward extension to other datasets without architectural changes. We therefore consider multi-dataset evaluation a natural and relevant direction for future work and explicitly highlight this aspect as extension opportunity for future work in the revised manuscript.

5. In Section IV, it is recommended to add a pseudo-code table. Additionally, the experimental results are currently presented only in tabular form; more intuitive and specific graphical representations are suggested.

→ We added a concise pseudo-code description to explicitly formalize the temporal execution order and decision logic of the proposed Tracking-by-Detection pipeline at the end of Section IV.

6. On page 4, line 3, "where" should be aligned to the left margin. Grammatical and spelling errors should be checked and corrected.

→ All indents after equations were removed. We also carefully checked the manuscript for grammatical and spelling error and made 31 changes only related to this point.

Reviewer #2: The manuscript proposes a modular MOT pipeline that keeps a Kalman-style backbone while adding three "tiny" neural modules—SPENT (prediction), SANT (single-measurement association), and MANTa (multi-measurement association)—with an emphasis on drop-in compatibility and embedded deployment. While the modularization premise is relevant, the current manuscript has multiple verifiable specification and methodology gaps that prevent the claims (including "KITTI tracking benchmark" performance) from being adequately supported.

1. The manuscript asserts drop-in compatibility with "standard Kalman-filter state updates," but does not provide a precise interface for uncertainty handling (e.g., how , , , , are obtained/propagated when SPENT replaces the prediction model);

→ In the proposed framework, SPENT replaces the deterministic motion model of the Kalman filter by directly predicting the next state. The uncertainty propagation is intentionally kept within the classical Kalman framework to preserve interpretability and compatibility. Specifically, SPENT outputs only the predicted state vector, while the state covariance $P_{k|k-1}$ is propagated using the standard Kalman prediction equation with a fixed process noise covariance Q. The measurement update, including the Kalman gain computation and covariance update, remains unchanged.

→ We have clarified the needed interfaces explicitly in the revised manuscript and added a structured description of inputs and outputs for SPENT and the association networks in Fig. 3 and Fig. 4 description.

2. Training/evaluation rely heavily on GT tracks; association labels are synthetically constructed (next-frame states + small noise), which avoids realistic MOT challenges (FP/FN, occlusion, ID switches);

→ We acknowledge the reviewer's concern regarding the use of ground-truth tracks and synthetically constructed association labels. Our intention in this work is to isolate and analyse individual MOT sub-tasks under controlled conditions, rather than to address the full complexity of end-to-end tracking.

→ The synthetic noise injection is explicitly designed to introduce controlled measurement uncertainty while avoiding confounding factors such as false positives, missed detections, or occlusions. This allows us to systematically evaluate the learning behaviour of SPENT, SANT, and MANTa. Additional explanations were added to Section IV-B.

→ We agree that handling false positives, false negatives, and occlusions is essential for real-world MOT systems. Extending the training and evaluation to include these effects is therefore identified as an important direction for future work and is now explicitly stated as a limitation of the current study.

3. Per-track softmax classification does not guarantee a feasible global assignment (multiple tracks can pick the same measurement);

→ We thank the reviewer for highlighting this important theoretical limitation. Indeed, the per-track softmax classification used in SANT / MANTa does not explicitly enforce a globally feasible one-to-one assignment. In practice, this situation does not occur in our experiments, as the training data is constructed such that each sensor object is associated with at most one track, and conflicting assignments are not present in the ground truth labels.

DHBW
Duale Hochschule
Baden-Württemberg
Stuttgart

Prof. Dr. Matthias Drüppel
Computer Science Department
Center for Artificial Intelligence

DHBW Stuttgart
Lerchenstraße 1
70174 Stuttgart
Telefon +49 711 1849 4511
matthias.drueppel@
dhbw-stuttgart.de

www.dhbw-stuttgart.de

→ Consequently, selecting the same measurement for multiple tracks would lead to an increased loss during training and is implicitly penalized by the learning process. We emphasize that SANT / MANTa is intentionally designed as a local, learned association module that replaces heuristic distance metrics, while global consistency is maintained at the system level by the surrounding Tracking-by-Detection framework.

4. The SPENT loss sign appears incorrect; padding is described inconsistently; MANTa loss form does not clearly match the stated one-hot setup;

→ This was well spotted. The sign of the loss function in equation (3) was indeed wrong and is now corrected.

→ We have re-written the one-hot vector explanation for MANTa in section IV-C to make it clearer which parts are one-hot encoded. We did not find an error in the written cost equation (5) but found that the explanation should be made clearer. We thus added a note explaining the connection between the matrix entries of the prediction and the loss function.

→ We clarified the padding strategy in the revised manuscript. All sequences are padded using zero-padding at the end of each sequence, and sequence lengths are sorted prior to mini-batch formation to minimize padding overhead. Padding tokens are masked implicitly by the LSTM through sequence length handling. We have updated the manuscript for clarification (Section IV. A.).

5. Missing key hyperparameters and training details;

→ We acknowledge this important concern and have added comprehensive training details to the revised manuscript. The configuration is now documented in a dedicated section (Section IV, c) overall training setup)

6. "Real-time/embedded" positioning lacks quantitative runtime evidence;

→ We thank the reviewer for highlighting this and added desktop median inference in section V. Experimental Evaluation.

7. Scholarly completeness issues: references and language.

→ We carefully went through the manuscript and improved it at various places. Grammatical and spelling corrections: 32 changes. New relevant references: 7 added.

We again thank both reviewers for their most valuable feedback that they provided voluntarily. We believe that the new changes largely improved our paper, especially in clarity and motivation for our work.

Sincerely,
Christian A. Holz, Christian Bader, Markus Enzweiler, Matthias Drüppel
Daimler Truck Research, Esslingen University of Applied Sciences, Corporative State University Stuttgart (DHBW)

(Prof. Dr. Matthias Drüppel)